

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

Introduction to Spatial Science and Technology

Time: 60 minutes

Total Marks: 30

Note: Put the question number and write the answer

Section: I (22 Marks)

1. What is radiometric resolution? Suppose you have a digital image which has a radiometric resolution of 6 bits. What is the maximum value of the digital number which could be represented in that image?
2. Define (a) Atmospheric blinds (b) Radiance
3. What is the basic difference between k-means and ISODATA classification approaches?
4. Differentiate between FOV and IFOV, what do they refer to in relation to satellite data?
5. What elements of image interpretation would you consider differentiating between (i) evergreen and mangrove vegetation (ii) forest from agriculture and plantations
6. You are assigned to extract water bodies of Hyderabad city using indices. Which index would you choose and why?
7. Explain the applications of shortwave infrared composite images
8. Draw the spectral reflectance curve of vegetation and explain the key factors responsible for its characteristic peaks and dips across different wavelengths?
9. How NDVI is used to monitor vegetation health?
10. List the applications of remote sensing in forestry
11. Give a brief account of dimensionality reduction technique discussed in the class that is used in satellite data processing?

Section-II (8 Marks)

12. Spectral response can be quite variable, even for the same target type, and can also vary with time, explain with an example?
13. Describe the geometric errors observed in the image taken by aircraft?